Austyn Nguyen

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Experience

Capital One McLean, VA
Product Management Intern *Jun. 2026 – Aug. 2026*

- Defined **UI, data-availability, and API requirements** with 4 engineering leads across 2 sprint teams — leading **sprint planning and backlog grooming** across the **SDLC** — launching **2 self-service products** serving **20M+** customers.
- Authored **PRDs** and **technical specifications** for Bank Statement modernization, translating system constraints into sprint-ready requirements and **OKRs** targeting **$10M** in projected savings and **$475K** annual OpEx reduction.
- Conducted **7-method customer research** (call listening, **28 customer/associate interviews**, competitive analysis across **26 services**) — mapping **3 customer journeys** and **4 behavioral modes** to drive **data-driven** product decisions.
- Spearheaded **competitive analysis** across **6 peer banks** (e.g., Wells Fargo, JPMC, Stripe) to design a **customer-facing help page** for VOD — surfacing UX patterns that shaped **go-to-market** strategy and **feature roadmap** prioritization.
- Coordinated **associate UAT pilot** across **125 test cases** covering navigation, PDF generation, and edge-case account rules — managing launch sequencing across **3 legacy teams** previously handling ~**1,000 manual calls** per 6 months.

Center for Healthcare Engineering & Patient Safety Ann Arbor, MI
Operations Research Assistant *Aug. 2025 – Apr. 2026*

- Partnered with Michigan Medicine's **Chief Transformation Officer** to model end-to-end patient flow across **inpatient and outpatient workflows** and prioritize a **$50M** capital portfolio using throughput and utilization metrics.
- Built **cost-impact and capacity-uplift models** to evaluate process-improvement scenarios, enabling **evidence-based** prioritization across efficiency, scalability, and ROI — directly informing the team's highest-impact strategic decisions.
- Gathered requirements from 3 clinicians to redesign **Epic EHR** scheduling logic — delivering **20+ automation scripts** that measurably improved **care-access equity** for underserved **walk-in patients** in a pediatric neurology clinic.

National Institute of Standards and Technology (NIST) Gaithersburg, MD
Software Engineering Intern *May 2024 – Aug. 2024*

- Owned the **product roadmap** for a data pipeline processing **500+ GB** across 27+ benchmarks; defined ingestion requirements, a **metadata schema standard**, and **API data contracts** with developers, eliminating manual curation.
- Translated complex pipeline architecture into a stakeholder presentation for **40 cross-disciplinary PhD scientists** — earning **organizational buy-in** for continued development and funding of the data pipeline roadmap.

Projects

ML Feedback Categorization Pipeline | *Capital One* May 2026 – Jul. 2026

- Designed a **two-tier text classification system** (phrase-match + ML) to auto-tag **472 monthly customer feedback comments**, freeing **6–7 hours/month** of manual work and enabling faster data-driven product iteration.
- Iterated across **5 ML approaches** (TF-IDF, BERT, LLM, hybrid); improved **precision from 67.2% to 86.4%** and cut false positives by **86.5%** — presenting tradeoffs and recommendations to non-technical stakeholders.

Agentic AI Workflow for Economic Impact Analysis | *Kite & Key Partners* Jan. 2026 – Apr. 2026

- Architected an **end-to-end agentic AI system** that autonomously researches and drafts economic impact analyses for **16 concurrent** state healthcare bills, entirely replacing a **time-intensive, manual process**.
- Delivered a **proof-of-concept** compressing analysis timelines **from weeks to hours**; co-authored a research paper on **LLM-driven** economic analysis validating technical feasibility with IHPI faculty **and opening policy access at scale**.

Day-Ahead Electricity Price Forecasting Product | *ElectricFish* Aug. 2025 – Dec. 2025

- Scoped and built a **3-stage hybrid ML pipeline** on **Azure ML Studio** to forecast **CAISO day-ahead electricity prices** across **10 California cities**, enabling engineers to implement a **dynamic pricing strategy** for EV stations.
- Validated against **3 years** of CAISO data via an **80/20** time-series split, hitting every threshold: $\mathbf{R^2 > 0.9}$, **RMSE** $<$ **10, sMAPE** $<$ **25** across cities, pricing zones, **and peak load periods**, meeting **client acceptance criteria**.

Education

University of Michigan – Ann Arbor Ann Arbor, MI
B.S.E. in Computer Science *Expected Dec. 2026*

- **Notable Coursework:** Project Management & Consulting, UI/UX, Agentic AI Software Engineering, Conversational AI, Machine Learning, Distributed Systems, Databases, Networks

Skills

Product: Product Strategy, Discovery, User Stories, Tech/API Specs, Business Requirements, System Design, Agile/Scrum, Roadmapping, OKRs/KPIs, User Research, Customer Journey, Stakeholder Management, Risk Mitigation, GTM
Technical: Python, SQL, Data Pipelines, Integration, Microsoft Azure, Azure ML Studio, Generative AI
Tools: Jira, Figma, Confluence, Power BI (Certified)
Campus Involvement: Pops Orchestra (Principal Violist), Chamber Ensemble, Zeta Pi Technical Fraternity